ANGELIA

SEMANTIC ARCHITECTURE

Adopted architecture • ownership • hand-offs
linguistic evidence • current achievements • challenges

Angelia (Ἀγγελία, *ang-geh-LEE-ah, hard g*)
Greek personification of messages and tidings; daughter of Hermes in Pindar.

01 LINGUISTIC CORE

02 EVENT & STORY GRAPH

03 EDITORIAL PRODUCTS

ARCHITECTURE STATUS: ADOPTED
12 named domains, descriptive Swift identifiers retained, maturity gates unchanged. Register of record: `docs/ANGELIA.md`.

PINNED REPOSITORY SNAPSHOT

Commit: `eb9f73f9205b4c9a32554567eeb1bd96349ea848` (head of `main`, 11 November 2026, 22:58 CET)
Repository: https://github.com/marspater/NewsApp-macOS
Register: `docs/ANGELIA.md` · Principal code: `Sources/Intelligence/Events/EventMatcher.swift`
Evidence: `docs/audits/2026-11-10-label-review.md`; issues/737

EDITION 16 Re-pinned after the Today edition and the name register landed on `main`. Today moves to live, the ownership register follows `docs/ANGELIA.md`, and the Today figures are re-read from the 11 November audits. Changes from edition 15 are listed on page 06.

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01 / EXECUTIVE ASSESSMENT

Architecture is strong. Research remains the bottleneck.

ANGELIA is the name for the integrated linguistic, semantic and editorial intelligence architecture of News for macOS. Angelia (Ἀγγελία) is the Greek personification of messages and tidings.

Executive finding. The main quality gap sits between accurate pair evidence, conservative event admission, and still-evolving occurrence labels. A larger model is not automatically the remedy.

STATUS	WHAT EXISTS / WAS MEASURED	CONSTRAINT
Live	Rules-based taxonomy; NaturalLanguage extraction; MiniLM and PairScorer; event and story grouping; hybrid search; source-cited overviews; the Today edition; Briefing; World Tension.	Correct pair scores do not guarantee correct full-group membership.
Measured	Topic holdout; label review; real-library replays; runtime and Neural Engine performance; perspective audit; Today shadow evaluation, its 11 November re-run, page timing and a first pass in a window.	Different samples and protocols have different denominators.
Experimental	Fact novelty, momentum, MMR ranking, wire identity and impression-backed personalization.	Implemented and measured; none has a caller in the app at the pinned commit.
Open research	Revised coverage objective; one accepted holdout; independent case adjudication; clustering-order effects.	A 60% multi-publisher target exceeds one sample's measured opportunity ceiling.

Four findings that frame the review

57% same-act co-location

86 of 151 reviewed same-occurrence pairs ended in the same event; 65 remained separated. [E01/E02]

13 admission refusals

Of 33 traced missed pairs, 13 were scored matches but refused by whole-event placement. [E03]

105 labels changed

51 of 400 Stage-1 and 54 of 300 validation first-pass labels changed after blind multi-reader review. [E02]

~38% feed-mix ceiling

On the 3 November library, estimated achievable multi-publisher share was ~38% (35–42%). [E03]

Architecture scope. ANGELIA spans Services, Intelligence and Storage, with owned files in Models and Coordinators. It is broader than the `Intelligence` source folder documented in `docs/ARCHITECTURE.md`, whose five-layer map stays the contract for technology and invariants; `docs/ANGELIA.md` is the name register. The protected networking gateway is an external security boundary, not an ANGELIA subsystem. [C01/C35]

Scope of this paper: source and dated-audit review pinned to the commit on the cover. No private evaluation dataset, live macOS build, on-device inference or user-local MLflow ledger was independently accessed. Issue #737 was not re-read for this edition.

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02 / SYSTEM IDENTITY

One system. Twelve accountable domains.

The Greek pantheon is an engineering vocabulary, not a fictional genealogy. ANGELIA is the integrated system; every named domain owns a concrete contract. Colour marks the layer: acquisition, storage, occurrence identity, evidence and editorial, research-gated.

IRIS feeds & wire identity	**METIS** linguistic signals	**PROTEUS** vectors & pair model	**THEMIS** pair verdict	**HARMONIA** event membership	**ARIADNE** story relations
MNEMOSYNE storage & search	**CLIO** sourced overviews	**ALETHEIA** fidelity & defence	**KAIROS** novelty & momentum	**ATHENA** editorial selection	**ERIS** World Tension

Named domains and exact responsibilities

DOMAIN	OWNS	TO / FROM
Iris	Feed fetching and parsing into storage; experimental wire identity.	Publisher records → Mnemosyne; origin groups → Kairos (R&D).
Metis	Classification; entity, action, date and language signals; enrichment; reader summary.	Structured `EventFeatures` → Themis, cached in Mnemosyne.
Proteus	MiniLM 384-d vectors and PairScorer `p(same)`.	Cosine + pair score → Themis; embeddings → Mnemosyne.
Themis	Hard conflict guards, rule assessment, pair verdicts, verdict calibration.	Pair assessment → Harmonia and Ariadne.
Harmonia	Admission, fragment merger, reader inclusions and exclusions.	Event membership → Mnemosyne.
Ariadne	Developing-story grouping, earlier-event relations.	`story_id` and related-act links → Mnemosyne.
Mnemosyne	SQLite provenance, FTS5, hybrid search and cached signals.	Persist and retrieve across the whole system.
Clio	Compose sourced overviews, passages, perspectives, impact and angles.	Evidence-backed overview → Aletheia → reader.
Aletheia	Claim checks, quality audit, prompt defence, generated-text fidelity.	Accept, reject or safe fallback → reader surfaces.
Kairos	Novelty and momentum research, currently gated.	Experimental facts and cadence → evaluation, not a live dependency.
Athena	Live Today edition, Briefing, importance and triage; experimental ranking and personalization.	Lane plan → Today page; prioritized stories → Briefing and banners.
Eris	World Tension assessment, trend and hotspot lenses.	Bounded geopolitical reading → sidebar, Briefing strip, sheet.

Owner comment rule. Each owned Swift file names its owner in `// Angelia · <owner>` one blank line after the license header. `script/format.sh --check` and the pre-commit hook enforce it under `Sources/Intelligence/`; owned files in Services, Storage, Models and Coordinators carry the same comment by the register. A name is an ownership alias and never raises maturity. [C41/C42]

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03 / IMPLEMENTATION REGISTER

Ownership register: named roles and code

The current architecture map for the pinned code. “Live” means called by the app; “experimental” means implemented under research or evaluation boundaries with no caller outside its own files and the test harness.

NAME	IMPLEMENTATION OWNER / IMPORTANT FILES	MATURITY
IRIS	`FeedFetcher`, `FeedXMLParser`, `JSONFeedParser`, `DateParser`, driven by `FeedManager`; `EditorialWireIdentity`.	Ingestion live; wire identity experimental.
METIS	`ArticleIntelligence` (`ArticleClassifier`, `NaturalLanguageTopicClassifier`); `EventFeatures`, `EventPlaces`; `ActClass`; `EnrichmentQueue`; `ContentExtractionPipeline` (reader text on open); `EntityIndexer`.	Live.
PROTEUS	`SentenceEmbedder.swift` (`SentenceEmbedder`, `PairScorer`, `BertWordPieceTokenizer`); `Resources/CoreML`.	Bundled models live.
THEMIS	`EventMatcher` v15, `EventMatchPolicy`, `EventConflict`, `EventPairAssessment`, `PairCalibration`.	Live. Calibration dormant below 50 reader verdicts.
HARMONIA	`EventClusterer` admission and `mergeFragments`; `event_inclusions`, `event_exclusions`.	Live.
ARIADNE	`EventClusterer.linkStories`; `EventStoryRelation`.	Live story association.
MNEMOSYNE	`DatabaseEngine*` (including `+Briefing`, `+Search`), `ArticleStore*`; `EventCandidateFinder`, `EventMatchKey`; `EntityGraph`; `MigrationCoordinator` †, `Models/EntityModels` †.	Live cross-cutting data layer.
CLIO	`OverviewComposer`, `OverviewGenerationCoordinator`, `OverviewPassageSelector`, `OverviewPerspectivesExtractor`, `OverviewTimelineBuilder`, `OverviewTimelineExtractor`, impact and thematic-angle extractors, `PassageFactExtractor`; `DualReaderAlignmentService`, `PerspectiveReports`; `Models/EventOverview` †, `Models/DualReaderModels` †.	Live sourced overview pipeline.
ALETHEIA	`OverviewClaimVerifier`, `GeneratedTextFidelity`, `OverviewQualityAuditor`, `CoverageSentimentEvaluator`, `GenerationPromptDefense`.	Live guardrails; agreement with the source, never external truth.
KAIROS	`EditorialNovelty`, `EditorialMomentum`.	Experimental; no production consumer.
ATHENA / LIVE	`Intelligence/Today/*` (`TodayEditionAssigner`, `TodayEditionLoader`, `TodayLane`); `FiniteBriefing`, `BriefingPriority`; `Intelligence/Briefing/*`; `StoryImportance`, `StoryCurator`; `NotificationService` triage.	Live. The app target rates no importance with a model.
ATHENA / EVAL	`EventJudge` (`EventJudgeReport`, the importance audit’s input).	Evaluation only; no production caller.
ATHENA / R&D	`EditorialRanking` (MMR); `EditorialPersonalization` (EWMA, impressions).	Experimental; no production consumer.
ERIS	`Intelligence/Tension/*`: `WorldTension`, `TensionMethodology`, `Hotspot*`, `WorldTensionNearYou`; `Models/HotspotModels` †.	World Tension live; not an external truth index.
ANGELIA / CORE	`NewsTextModel`, `AppleIntelligenceAvailability`, `IntelligencePolicy`.	Shared model, availability and energy service.
NO SINGLE OWNER	`Coordinators/FeedManager`: starts the refresh (Iris), background classification (Metis) and the clustering pass (Harmonia and Ariadne); also owns sleep, wake and lifecycle.	Live; its comment names the domains it drives.

Boundary rule. The protected network gateway (`SecureHTTPClient`, `NetworkBoundaryProxy`, `IPAddressValidator`) handles hardened transport, DNS constraints and Web preview isolation. It feeds source acquisition but does not become a myth-named semantic owner. [C01/C35]

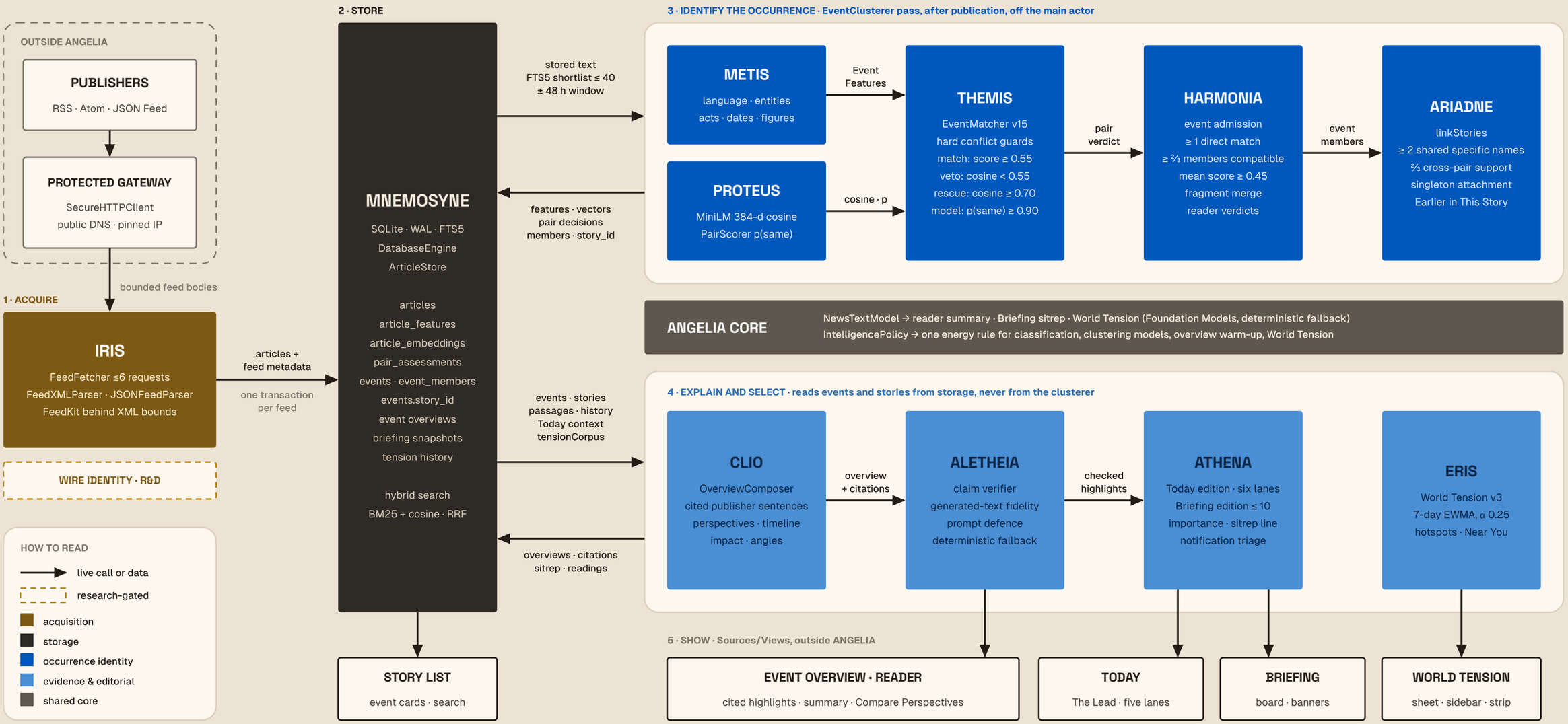
† Ownership recorded on branch `mars/angelia-unification` (`841e3c4`), not yet on `main` at the pinned commit; the files themselves exist on `main`. Views sit outside ANGELIA and belong to no domain. Authority rule: Aletheia validates textual support; Themis judges report pairs; Harmonia controls group membership; Athena controls placement. No subsystem may silently promote another system’s evidence into a stronger claim.

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04 / WHOLE-SYSTEM VIEW

ANGELIA | System map

Read it left to right, then top to bottom: acquire, store, identify the occurrence, explain and select, show. Mnemosyne is the only bridge between the lanes. The clusterer writes memberships to storage and every editorial surface reads them back from there, so no arrow crosses another.



Gateway and views sit outside ANGELIA. Eris feeds a Briefing strip for display only and takes no part in story selection. Thresholds shown are the `EventMatchPolicy` defaults at the pinned commit.

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05 / DATA CONTRACTS

ANGELIA | Operational hand-offs

Every row names the producing owner, the payload and the receiving owner. Rows run in pipeline order, so the page reads as a ledger rather than a wiring diagram.

ACQUISITION → STORAGE → OCCURRENCE IDENTITY

PUBLISHERS	feed bodies through the protected gateway	IRIS
IRIS	articles + feed metadata · one transaction per feed	MNEMOSYNE
MNEMOSYNE	`matchingText` : headline + first 600 description characters	METIS
MNEMOSYNE	the same text, for the sentence embedding	PROTEUS
MNEMOSYNE	shortlist: ≤ 8 FTS5 terms · ≤ 80 rows read · ≤ 40 kept · ± 48 h	THEMIS
METIS	`EventFeatures` · who / what / where / when · `ActClass`	THEMIS
PROTEUS	cosine, and `p(same)` for borderline pairs	THEMIS
THEMIS	`EventPairAssessment` + conflict	HARMONIA
HARMONIA	formed events; the story pass applies the pair rules alone	ARIADNE
HARMONIA	event IDs · members · membership version	MNEMOSYNE
ARIADNE	`story_id` per active event · related-act rows	MNEMOSYNE

STORAGE → EVIDENCE AND EDITORIAL → SURFACES

MNEMOSYNE	events · members · stored passages	CLIO
CLIO	composed overview + per-source citations	ALETHEIA
ALETHEIA	accepted text or deterministic fallback	OVERVIEW · READER
MNEMOSYNE	≤ 500 candidates · importance · coverage · reading history	ATHENA
CLIO	cited highlights of the top threads, read from storage	ATHENA
MNEMOSYNE	assignment context of the listed stories (`TodayEditionLoader`)	ATHENA
ATHENA	lane plan from `TodayEditionAssigner.assign`	TODAY PAGE
ATHENA	frozen daily edition ≤ 10 · sitrep lines	BRIEFING
ATHENA	priority · starred · normal banner lanes	NOTIFICATIONS
MNEMOSYNE	`tensionCorpus` day-panel headlines	ERIS
ERIS	`WorldTensionState` · reading · drivers · hotspots (display only)	WORLD TENSION
ANGELIA CORE	model access and one outcome mapping	3 PROSE SURFACES
ANGELIA CORE	energy rule: Low Power Mode, serious thermal state	BACKGROUND WORK

IRIS · WIRE ---- origin and wire groups, offline --▶ KAIROS

KAIROS -------- novelty · momentum ------▶ EVALUATION

ATHENA · RANKING ---- MMR order · personalization --▶ EVALUATION

Shared source files, split ownership

FILE	OWNER A	OWNER B
`EventMatcher.swift`	Metis · `EventFeatures`, `EventPlaces`	Themis · `EventMatcher`, `EventMatchPolicy`, `EventConflict`, `PairCalibration`
`EventClustering.swift`	Harmonia · admission and fragment merging	Ariadne · `linkStories` and story components
`EventCandidates.swift`	Mnemosyne · `EventCandidateFinder`, `EventMatchKey`	Ariadne · `EventStoryRelation` and earlier links
`ArticleIntelligence.swift`	Metis · classification and reader summary	Core · `NewsTextModel`

Shared storage rule: Iris writes articles first; Mnemosyne supplies retrieval and stored context to event matching, Clio, Athena and Eris.

Changed from edition 15

- **Today is live.** `ArticleListView` shows `TodayEditionView`, and `Views/Today/TodayEdition.swift` calls `TodayEditionAssigner.assign`. Edition 15 was pinned to `ba1853e866fd`, before the views landed (#1222 to #1226); its “not live” correction is withdrawn.
- **Names are in the repository.** `docs/ANGELIA.md` is the register; each owned Swift file carries a `// Angelia · <owner>` comment and `script/format.sh --check` enforces it under `Sources/Intelligence/`.
- **Register follows `docs/ANGELIA.md`.** Views and `DatabaseEngine+Briefing` left Athena; `EventJudge` (evaluation only), `CoverageSentimentEvaluator`, `PassageFactExtractor`, `OverviewTimelineExtractor`, `EventMatchKey`, `EventMatchPolicy`, `EventConflict` and the model files were added; `FeedManager` is listed with no single owner.
- **Role fixes.** `ActClass.swift` is Metis; `BriefingSitrepComposer` is Athena; `SentenceEmbedder.swift` is wholly Proteus and left the shared-files table.
- **Today figures.** Page 14 now reports the 11 November re-run with the reader’s real subscriptions and corrected labels, the page timing and the first pass in a window.
- **Re-pinned** to `eb9f73f9205b`. Model-file owners and the tests table are on the unmerged branch `mars/angelia-unification`.
- **Standing from edition 15.** Iris ingestion files, shared-core clients, the 80-read / 40-kept shortlist, and editorial reads through storage.

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06 / LINGUISTIC ONTOLOGY

Different wording is not the same as a different event

A report is an article; an event is a single occurrence (actors, deed, place, time); a story is a linked sequence of acts; a topic is a broader common subject. These entities cannot be collapsed into “similar articles.”

SEMANTIC TYPE	IDENTITY RULE	CORRECT PRODUCT HANDLING
same	Both reports describe one act at a compatible time and place with consistent participants.	One event. May have multiple reporting publishers.
pegged-part	A contemporary scene or detail of the act; strict inclusion only after individual verification.	Possible event inclusion only if verified.
pegged-reaction	Words spoken about the act, without reporting a new deed.	Contextual reading beside the event; not a strict event positive.
pegged-analysis	Explainer or analysis about the act, without a new deed.	Contextual reading; not automatically a member.
later	Subsequent response, legal action, decision, continuation or consequence constituting a distinct act.	Separate event, possibly same developing story.
topic / unrelated	Only a general subject is shared, or nothing meaningful connects the reports.	Neither occurrence merge nor unsupported story link.

Figure 3 · occurrence boundary



Adopted rule (10 Nov). Only `same`, and individually verified contemporaneous `pegged-part`, count as strict event positives. Reactions and explainers may accompany a story without voting in admission; `later` stays a distinct act. [E02]

Why the labels are expensive

- “Three killed” and “five killed” can be updates to one attack, not separate acts; casualty tolls are treated specially.
- Two protests on successive days may share the same country, actors and vocabulary while being different occurrences.
- Two reports can be topically dissimilar yet describe one event; a paraphrase embedding should help, not define the label.
- Announcements, previews, court decisions, live blogs, reactions and explainers need explicit boundary examples.

Historical labeling protocol: six rubric gaps remain (linked reactions and explainers; announced vs completed act; election phases; day-two disaster coverage; short multi-day waves; explainers that also report the act). Five three-way splits still require review. [E02]

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07 / LINGUISTIC FRONT END

Metis: language, entities, action and category

Metis builds a lightweight, deterministic semantic fingerprint from the headline and the first 600 characters of the description, using Apple NaturalLanguage and custom rules. The system does not assume full-article text exists.

SIGNAL	WHY IT EXISTS	KNOWN FAILURE
People / organizations / places	Specific anchors distinguish actors and locations; demonyms and country aliases normalize references. Headline places and people are kept apart from the description's (v15).	Common names or absent NER signals; a country alone does not imply occurrence.
Title case and untyped names	Recover headline names that system tagging misses.	Capitalization is language and publisher dependent.
Lemmas and `ActClass`	Distinguish resignation, vote, attack, arrest, sanction and report consequences.	Mixed clauses, ambiguous cue words, indirect speech.
Date / time / weekday	Constrain occurrence identity across near reports.	Incorrect feed dates, retrospective pieces and "day two" updates.
Numbers / casualties	Different headline figures can separate incidents; mutable tolls stay distinct.	Toll revisions and generic statistics can mislead matching.
Source series / edition	Remove repeated newsletter prefixes as evidence and distinguish morning and evening slots.	Newsletter items can describe multiple acts.
Language eligibility	Avoid English-only neural inference for recognized non-English text.	Unrecognized language still passes English neural eligibility.
RSS category + weighted cues	Twelve categories; feed label first, headline-biased lexical scoring, World fallback.	Editorial conventions; vocabulary coverage and publisher shift.

Measured 12-topic classification (model-annotated data)



Evaluation reading. On a fresh 200-report sample, classification increased from 113/200 (56.5%) to 142/200 (71.0%). The 182/200 (91%) figure comes from the tuned development sample and is not deployment accuracy. Labels came from Claude with same-family review, not independent human gold. [E05]

Model-text boundary. Metis also owns `EnrichmentQueue` and source-content feature extraction. It does not use a generative model during ingestion classification. The app calls a text model on three bounded prose surfaces only (reader summary, World Tension, Briefing sitrep); those are coordinated by ANGELIA's shared core and checked by Aletheia. [C02/C03/C23]

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08 / SEMANTIC REPRESENTATION

Proteus: two models, two different questions

Embeddings retrieve meaning-near candidates cheaply; pairwise cross-encoding directly evaluates an eligible report pair. Neither determines whether a full group forms one occurrence.

MODEL	CONFIGURATION	PRODUCES / USED BY
MiniLM (bi-encoder)	`all-MiniLM-L6-v2`; WordPiece; 384-d L2-normalized vectors; 128-token maximum; fixed functions 16/32/64/128.	Cosine for Themis; vectors stored by Mnemosyne; hybrid search, novelty and ranking providers.
PairScorer (cross-encoder)	MiniLM-derived shipped binary model; pair input up to 256 tokens; fixed 64/128/256; `.cpuAndNeuralEngine`.	Direct `p(same)` on conflict-free borderline pairs. A failed three-class retrain did not replace shipped weights.
Foundation Models	Apple's local system text model; depends on OS, hardware, language availability and user control.	Reader summaries, World Tension explanation and Briefing sitrep only; bounded, with deterministic fallback.

EMBEDDING / MiniLM

Text A → vector A; text B → vector B. `cosine(A,B)` measures proximity of representations, not calibrated event probability.

PAIR SCORER

Both texts in one model → `p(same)`. Eligible only after rule and language checks, and only when borderline.

Current default thresholds. Semantic divergence veto below cosine 0.55; eligible paraphrase rescue at cosine ≥ 0.70 with a specific anchor; PairScorer cutoff 0.90. These are policy choices tuned to particular datasets, not universal probabilities. [C05/C07]

OPTIMIZATION	DOCUMENTED RESULT	INTERPRETATION
Fixed-size model functions	2,000-text batch ~927 → 742 ms on an Apple M5 corpus-text benchmark.	ANE placement and batch mechanics matter more than model-size speculation.
Candidate language reuse	Warm from-scratch re-clustering 55.65 → 16.21 s; cold 83.36 → 16.96 s on 1,454 rows.	Same gate outcomes on the measured copy; machine load differed.
Model-version protection	Embeddings are stamped and invalidated when inputs or version change.	Prevents cross-version vector comparison and stale cache reuse.

Recognized non-English report pairs skip neural evidence; an unknown language is eligible under current code. Energy limitations defer model-dependent decisions rather than silently declaring final negative results. [C02/C05/C06]

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09 / MATCHING AND ADMISSION

Themis decides pairs; Harmonia decides groups

Keeping these as separate decisions is central to understanding current recall loss. Pair compatibility is not an equivalence relation, and one member can block a plausible new join.

Figure 5 · semantic decision pipeline

1. MNEMOSYNE / CANDIDATES
FTS5 shortlist within ±48 hours, up to 40 rows kept; 72-hour active-event lifetime. Recall of candidates is not cluster recall.

2. METIS + PROTEUS / SIGNALS
Linguistic anchors, action and date cues, cosine if eligible, borderline `p(same)`.

3. THEMIS / PAIR ASSESS
Hard conflicts; lexical, anchor and time score; semantic veto and rescue; confirm borderline when permitted.

4. HARMONIA / ADMISSION
At least one direct match, ≥2/3 compatible, all-member evidence, no blocking conflict, adequate mean score.

Core deterministic equation.
`score = 0.4 × lexical overlap + 0.4 × entity/anchor overlap + 0.2 × time proximity`. Standard direct-match floor: 0.55. Pair conflict and specific-anchor requirements still govern admission. [C05]

STAGE / POLICY	DEFAULT	REASON
Pair time gap	36 h maximum; 12 h strict; full credit through 6 h.	Close publication alone does not imply one act.
Semantic veto / rescue	Cosine < 0.55 veto; ≥ 0.70 may rescue.	Paraphrase help cannot erase known contradictions.
PairScorer	`p(same)` ≥ 0.90 for eligible borderline pairs.	Expensive contextual analysis applied sparingly.
Verdict calibration	`PairCalibration` moves the cutoff by at most ±0.05; dormant below 50 reader verdicts.	Reader corrections adjust the model's cutoff, never the rules.
Event membership	≥1 match; ≥2/3 compatible pairs; mean score ≥ 0.45; maximum 100 members.	A single incompatible member can refuse admission.
Experiment knobs	Country-only signal, veto-share, anchorless model, story model and pair-model prefix are default-off.	Out-of-sample and false-link gates have not been passed.

A reader's explicit "Same Event" inclusion and exclusions affect grouping; matcher changes revalidate active members, not the entire archive. Pair decisions are stored as audit rows. [C05/C06]

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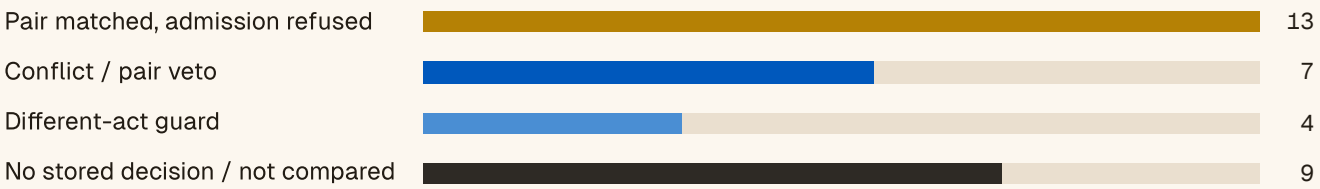
10 / GROUPING MECHANICS

Correct local pairs can still form incorrect cards

Harmonia operates incrementally: it tests a report against every member of candidate events, then uses fragment merging; Ariadne separately forms developing-story links between distinct events.

GROUPING STAGE	QUESTION	FAILURE MODE
Pair judgment / Themis	Do A and B report the same occurrence?	Missed paraphrase or false shared-actor match.
Admission / Harmonia	Can A fit every member of event E?	Third member vetoes a good A–B pair.
Fragment merge / Harmonia	Should E1 and E2 be one occurrence?	Two parts of same act remain duplicate event cards.
Story pass / Ariadne	Are distinct events causally related developments?	Topic-only connections, or later acts leaking into same event.
Display / Athena & Clio	Which story is shown with what context?	Wrong groups inflate source breadth, perspectives and importance.

Where 33 missed same-act pairs stopped



Finding. 13 of 33 missed Stage-1 pairs were directly scored “match” but rejected by whole-group admission. In 12 of those 13, members in the other event raised vetoes; ten of twelve also failed broader agreement criteria. Candidate search found 99 of 100 same-labelled pairs, and no miss was below the pair-model cutoff. [E03]

Why arbitrary threshold relaxation stalled. The 6 November setting g improved the 3 November sample but failed out of time: story links on validation declined 47 → 43. The correct next comparison is a frozen-label, arrival-order and membership-partition replay, not more ad-hoc veto exceptions. [E01/E04]

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11 / RETRIEVAL AND PROVENANCE

Mnemosyne owns search; Proteus supplies vectors

Memory is cross-cutting. ANGELIA writes features, embeddings, version stamps, user corrections, event membership, pair assessments and sourced output to local storage.

SEARCH / DATA PATH	CURRENT IMPLEMENTATION	SEPARATION OF CONCERNS
Article shortlist for event matching	`EventCandidateFinder` uses title and description terms (≤ 8 FTS5 phrases), publisher timestamp window and row cap.	A candidate is a possible comparator, not a semantic verdict.
Hybrid archive search	SQLite FTS5 BM25 plus MiniLM cosine arm; reciprocal rank fusion (RRF k=60) and cosine floor 0.35.	Proteus creates vectors; Mnemosyne executes retrieval and ranking.
Cached linguistic signals	`article_features` versioned with `EventFeatures.storageVersion` (5); stable article IDs.	Edits invalidate relevant features instead of preserving stale judgments.
Pair provenance	`pair_assessments` stores result, score and numeric features.	A recorded score does not by itself explain all final membership decisions.
Entity indexing & graph	Metis creates indexes; Mnemosyne stores the graph and relations.	Indexing is not fact verification.
Output caching	Event overview inputs, membership and version stamps, and fallback handling.	Changed source or membership invalidates old compositions.

Data ownership map

WRITE / PROVENANCE

Metis: linguistic features. Proteus: embeddings. Themis: pair evidence. Harmonia and Ariadne: memberships. Clio and Aletheia: sourced overview artefacts.

READ / CONSUMER

Mnemosyne: centralized persistence and query behavior. Story list, Briefing, Reader and research harnesses consume stable records.

Operational boundary. RSS, Atom and JSON Feed acquisition and secure HTTP ingestion remain in Iris and the protected gateway respectively. Network transport is not a linguistic trust guarantee. The gateway is outside ANGELIA; publisher text remains untrusted even if fetched safely. [C01/C35]

Search evaluation is not event evaluation. Search should be judged on relevance, abstention, query coverage and ranking. Event grouping is instead judged on same-occurrence precision and recall, duplicate cards, false joins and order sensitivity. High-quality semantic search cannot be used as proof of accurate clustering. [C08]

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12 / OVERVIEWS AND CONFIDENCE

Clio composes; Aletheia checks what is supported

Source attribution and factual truth are different things. The live overview path prioritizes publisher excerpts, selected passages, perspectives, timeline cues, industry impact and thematic angles.

OWNER / PATH	PRESENT BEHAVIOR	QUALITY CONTRACT
Clio / passages	`OverviewPassageSelector`, `OverviewComposer`, `OverviewGenerationCoordinator`.	Select source-preserved highlights with clear passage identity.
Clio / perspectives	`OverviewPerspectivesExtractor`, `PerspectiveReports`, `DualReaderAlignmentService`.	Speaker attribution must stay attached to the precise claim.
Clio / context	Timeline extraction and build; `OverviewIndustryImpactExtractor`; `OverviewThematicAngleExtractor`.	A development's time is not necessarily its article publication time.
Aletheia / checks	`OverviewClaimVerifier`, `OverviewQualityAuditor`, `GeneratedTextFidelity`, `GenerationPromptDefense`.	Reject unsupported changes to numbers, action, negation and speaker.
Angelia / model	`NewsTextModel` with availability and energy gating.	Reader summary, World Tension and sitrep only; local fallback.

Measured source-fidelity results

69.6% two-voice coverage

Final 26 October live run: 80 of 115 multi-source events had at least two extracted perspectives; window-specific.

10 / 10 authored controls

Verifier fixes gave expected accept/reject outcomes across ten designed test cases; not general open-domain accuracy.

72 reviewed perspective rows

Final agent-led check found 72/72 selected changed rows correct; small and not independent human truth.

No generated event overview facts

The production overview uses source excerpts. Model-authored factual event introductions are not the live path.

Definition of Aletheia. "Verified" means supported by the cited available passage under deterministic rules, not that the publisher's claim is externally true, complete or fair. This is a source-fidelity layer, not a general fact-checking authority. [C02/C09/C11/C12/C22/E06]

Controlled generation surface. Where Apple Foundation Models are ready, ANGELIA's core can draft reader summaries, World Tension assessments and short executive sitrep lines. Aletheia's fidelity gates and deterministic fallbacks prevent unsupported prose from being treated as a required part of the event data model. [C02/C03/C32]

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13 / EDITORIAL PRODUCTS AND RESEARCH

Today and Briefing are live; Kairos is not yet

A subsystem can exist, pass deterministic tests and still have no approved production consumer. The Today edition crossed that line on 11 November: its six lanes are assigned by Athena and shown by the Today page. It shipped ahead of the natural-news evaluation listed under P2.

MYTH / SUBSYSTEM	IMPLEMENTED BEHAVIOR	STATUS / EVIDENCE GAP
Athena / Today	Six lanes: The Lead, In Motion, The Deep Read, Your Beat, Outside Your Usual, The Latest; deterministic scores over formed stories, assigned off the main actor.	Live. Thresholds still rest on one library copy and model-family labels with Mars's spot-check; In Motion and Your Beat variants are compared, not decided.
Athena / Briefing	Finite daily edition, publisher caps, importance, recency, diversity and reader category history.	Live; not the experimental MMR selector.
Athena / triage	`StoryImportance`, `StoryCurator` lead images, notification importance and triage.	Live; not an independent factual classifier.
Athena / MMR	`EditorialRanking`: semantic diversity with time decay among up to 500 stories.	Experimental: synthetic NDCG unchanged at 0.128571; median 74.67 ms Debug.
Athena / personalization	Opt-in impressions, clicks and saves, EWMA vector, bounded discovery allocation.	Experimental; no independent real-news relevance acceptance.
Kairos / novelty	Historical same-event retrieval; entity-anchored passage; MiniLM repetition screen.	Experimental; authored holdout 5 TP / 0 FP / 3 FN; no consumer.
Kairos / momentum	Report-volume and cadence replay, independent origin floors, acceleration and deceleration.	Experimental; depends on trustworthy wire and source identities.
Iris / wire identity	Agency credit, near-copy group and explicit unknown classification.	Experimental: 591/811 (72.9%) replay rows had unknown origin or identity.

What the Today evidence says (10 and 11 November)

MEASURE	RESULT	READING
Lead precision@5	4/5 for publisher count, Briefing signals and the Z2 score alike (28 real subscriptions, corrected labels); 5/5 on the first run.	Five places on one day do not separate the three rankings.
In Motion	Shows 5 stories: 4 real updates, 1 borderline, no repetition. Misses 6 of 10 real updates.	All 6 are kept out by the rising-momentum rule; dropping it admits 6 of 7 repetitions.
Your Beat	229 of 406 daily stories (56%) qualify before the cap of 12; 254/522 (49%) on the first run.	Publisher affinity favours high-volume feeds; source volume can masquerade as relevance.
Stored full body	159 of 738 stories (21.5%), first run.	Deep read, novelty and wire identity are content-limited.
Page assignment	200-story page: context load about 41 ms, assignment 727 → 125 ms median after spreading stories over cores.	Measured on a busy Mac; 250 ms proposed as the budget.
First pass in a window	macOS 27.2.1 only. First card about 880 ms on a quiet Mac against an 850 ms budget; peak memory about 130 MB above main.	None of the three Z5 budgets is recorded as met; macOS 26 and 15, trackpad and VoiceOver not run.

Causal dependency. A false event merge can simultaneously inflate apparent source diversity, momentum, viewpoint coverage and editorial urgency. Improve grouping and origin confidence before trusting sophisticated ranking outcomes. [E08/E09/E10/E11/E17/E18/E19]

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14 / BOUNDED GEOPOLITICAL SEMANTICS

Eris owns World Tension and hotspots

Eris, the Greek personification of strife, is the name of the distinct geopolitical-assessment domain. It must never be confused with factual event matching or external verification of threats.

ERIS COMPONENT / CONCEPT	BEHAVIOR	TRUST LIMIT
`WorldTension`, `TensionMethodology`	On-device assessment of international news with four cited dimensions.	News-headline assessment is not an independently verified global-danger statistic.
Novelty	How unusual the assessed signal is relative to ordinary reporting.	Not the same as Kairos article-level new-fact precision.
Proximity to worst case	How close a situation appears to an extreme adverse outcome.	A model's judgement is bounded, not predictive certainty.
Reversibility	How easily the assessed harm could be undone.	Context-sensitive, requires readable explanation.
Scope of impact	Breadth of people, regions and systems potentially affected.	Headline and feed composition constrains its inference.
History & hotspots	7-day reading, rising-trend bonus, hotspot gazetteer and projections.	Map position and regional lenses do not establish incident truth.
Near You (`WorldTensionNearYou`)	Where the Mac's Region names a country next to a war, the past week's own-library stories add up to 15°; keyword counts, never the model.	A regional lens on the reader's library; the world score stays beside it.

World Tension v3 presentation mechanics

HISTORICAL READING

Seven-day EWMA of accepted daily assessments, alpha = 0.25. A consecutive-rise bonus adds up to 9 points. Missing days are skipped.

ESTIMATE BOUNDARY

Keyword fallback estimates are displayed distinctly and excluded from the accepted-assessment average. Stored history keeps up to 90 days.

> **Distinguish the questions.** Themis asks "is this the same occurrence?" Harmonia asks "does it fit the existing event?" Athena asks "where should this story appear?" Eris asks "what is the assessed global tension signal?" These outputs cannot share one quality metric. [C20/C21]

Operational and ethical guardrails

- Show cited explanatory factors and distinguish accepted assessment from fallback estimate.
- Treat geographic proximity and hotspot names as context, not as proof of an event or risk to a person.
- Measure sensitivity to dominant publishers, duplicated wires, language coverage and missing local reporting.

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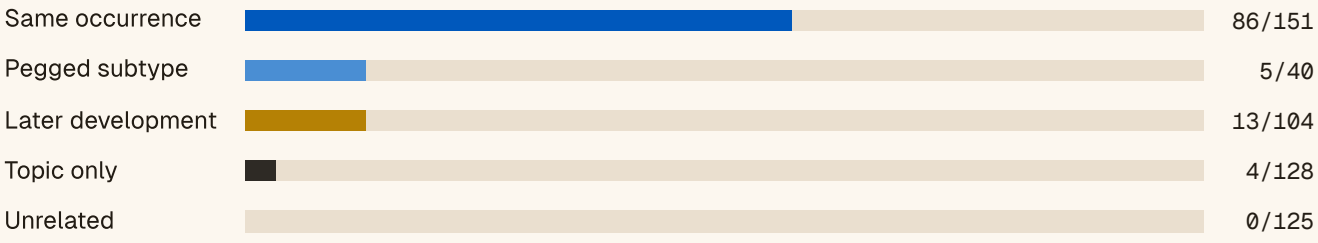
15 / EVIDENCE AND ERROR BUDGETS

Measured achievements versus unresolved deficits

The numbers below come from different dated audits. They are not interchangeable estimates of a single global accuracy rate. Dataset provenance is part of the measurement.

AREA	WHAT IS ACHIEVED	OPEN LIMITATION
Topic classification	Fresh labelled sample: 56.5% → 71.0%.	Label source and narrow feed/time sample.
Candidate retrieval	99/100 same-pair candidates found in one Stage-1 study.	Retrieval reach is not final event recall.
Event membership	86/151 reviewed same-act pairs in one event (57%).	65 split; 13 later and 4 topic pairs placed together.
Publisher opportunity	Estimated ideal multi-publisher share ~38% on 3 Nov mix.	Not a transferable ceiling for every subscription set.
Perspectives	80/115 events (69.6%) had ≥2 voices on final audited run.	Window, available text and reviewer-family limits.
Novelty	5 TP, 0 FP, 3 FN authored holdout.	Small synthetic experiment; not deployed.
Reclustering speed	55.65→16.21 s warm; 83.36→16.96 s cold.	Same tested copy; fluctuating host load.
Neural placement	MiniLM 2,000 corpus texts 927→742 ms.	Specific Apple M5 test conditions.

Figure 6 · reviewed pair membership, by class



Share of sampled labelled report pairs the baseline placed in the same event, including erroneously linked later and topic cases.

What the chart means. It is not a global precision estimate; different labels and replay periods change denominators. [E01/E02]

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16 / RESEARCH INTEGRITY

The measurement target must be better than the model

The November research demonstrates a genuine annotation and KPI problem. ANGELIA's next quality advance begins with definitions, opportunity-adjusted outcomes and independently checkable replays.

KNOWN RESEARCH ISSUE	OBSERVED EVIDENCE	REQUIRED CORRECTION
Label disagreement	51/400 Stage 1 and 54/300 validation labels changed after blind review.	Version the rubric; review disagreements; preserve original labels.
Model-family review dependence	All three blind readers came from the same model family; Mars has not adjudicated the disputed set.	Human or genuinely independent expert spot-check for difficult and critical cases.
Ambiguous `pegged` class	Reaction, analysis and part were previously mixed into later/same.	Apply the 10 Nov strict policy and verify individual pegged-part pairs.
Unattainable KPI	60% multi-publisher objective exceeded ~38% estimated ceiling for one library.	Opportunity-adjusted KPI across multiple feed and time mixtures.
Narrow old holdouts	3 Nov borderline sample cosine 0.55–0.70 weakly separated labels (AUC ~0.61).	Stratified pair sets and sealed independent acceptance split.
Greedy grouping confound	Changed thresholds alter event order and may lose previously correct joins.	Record arrival order and stage-by-stage veto attribution.
Source independence uncertainty	72.9% of wire replay rows had unknown identity.	Calibrate unknown vs independent; do not count copies as voices.

Research loop already in use, with needed additions

PRESENT / TOOLING

MLflow experiment ledger, marimo comparison notebook, Datasette read-only copies; commit, dataset and knob measures recorded locally.

NEXT / QUALITY CONTRACT

Add rubric SHA, independent adjudicator, event- and copy-disjoint splits, arrival-order seed and direct-pair vs final-group error fields.

> **Avoid repeating failed settings.** Country-only linking, g veto-share tuning, post-hoc evidence-head re-interpretation and premature three-way retraining already failed or confounded acceptance. The issue #737 "do not repeat" record is binding research context. [E01/E04/E16]

Source provenance on paper: every cited code path is pinned to commit eb9f73f9205b4c9a32554567eeb1bd96349ea848. Example: `Sources/Intelligence/Events/EventMatcher.swift`; experiment: `docs/audits/2026-11-10-label-review.md`. The repository is https://github.com/marspater/NewsApp-macOS.

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17 / RESEARCH PRIORITIES

The action sequence: contracts before new models

A stronger research architecture is an operational acceptance loop around the existing system, not a second implementation of the same semantic problem.

ORDER	WORK PACKAGE	PROMOTION EVIDENCE
P0 / now	Freeze one-occurrence policy: six rubric gaps, verified pegged-part decisions, small case pack and reviewer sign-off.	Named positives and negatives, recorded disputes and versioned label files.
P0 / now	Replace blanket 60% multi-publisher share with opportunity-adjusted coverage on multiple source mixes.	Measured ceiling, confidence intervals, independent multi-publisher opportunity.
P1	Instrument candidate → pair → each veto → member admission → fragment and story outcome.	Every sample split or false merge attributable to one stage.
P1	Replay multiple arrival orders and prior cluster partitions under fixed rules.	Reduced missed same-act pairs without higher wrong-event grouping.
P2	Re-evaluate constrained event- and act-aware representations only after label and grouping gates.	Frozen model, out-of-time event-disjoint labels and a sealed untouched test.
P2	Evaluate Athena's Today lanes and Kairos on natural news streams with defensible wire identity. Today shipped on 11 November ahead of this gate, so for Today it is now a post-release check.	Improved source-independent utility and natural-news metrics.
P3	Promote only with native runtime, memory, energy, fallback and rollback measurements.	Reproducible production acceptance and user-facing safety checks.

Six-gate acceptance flow

ONTOLOGY	LABELS	PAIR	CLUSTER	PRODUCT	PROMOTE
one act vs later	blind review	rules / ML	order + veto	card utility	holdout + cost

Recommended next experiment. With thresholds fixed, replay annotated same-act cases while recording every member veto and arrival-order permutation. Compare duplicate-card reduction and false-merge severity. This isolates the largest measured loss without claiming an unearned new model breakthrough. [E03/C06]

Presentation-ready message. ANGELIA combines deterministic linguistic cues, local semantic representations, pairwise identity arbitration, event assembly, source-backed explanation and live editorial selection. Its engineering foundation is established. Its principal open problem is reliable semantic definition and evaluation at event-group level, not raw inference speed.

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APPENDIX A / SOURCE REGISTER

Code ownership references: C01–C18

All paths are pinned to the exact code snapshot. The repository prefix is printed once; each row prints its path so the record survives offline printing.

Pin. Repository github.com/marspater/NewsApp-macOS • branch `main` • commit `eb9f73f9205b4c9a32554567eeb1bd96349ea848` • 11 November 2026. Printed paths append to `https://github.com/marspater/NewsApp-macOS/blob/eb9f73f9205b…/`. Every path below was checked to exist at that commit.

REF	ROLE	PRINTED PATH
C01	README / release scope	`README.md`
C02	On-device intelligence design	`Sources/News.docc/OnDeviceIntelligence.md`
C03	Metis / common generation wrapper	`Sources/Intelligence/ArticleIntelligence.swift`
C04	Mnemosyne / Ariadne shortlist	`Sources/Intelligence/Events/EventCandidates.swift`
C05	Metis / Themis core	`Sources/Intelligence/Events/EventMatcher.swift`
C06	Harmonia / Ariadne core	`Sources/Intelligence/Events/EventClustering.swift`
C07	Proteus / MiniLM & pair model	`Sources/Intelligence/Events/SentenceEmbedder.swift`
C08	Mnemosyne / hybrid search	`Sources/Storage/DatabaseEngine+Search.swift`
C09	Clio / overview composer	`Sources/Intelligence/Overview/OverviewComposer.swift`
C10	Clio / passages	`Sources/Intelligence/Overview/OverviewPassageSelector.swift`
C11	Clio / perspectives	`Sources/Intelligence/Overview/OverviewPerspectivesExtractor.swift`
C12	Aletheia / claim verifier	`Sources/Intelligence/Overview/OverviewClaimVerifier.swift`
C13	Proteus / pair metadata	`Sources/Resources/CoreML/pairscorer-metadata.json`
C14	Proteus / MiniLM metadata	`Sources/Resources/CoreML/minilm-metadata.json`
C15	Athena / Today	`Sources/Intelligence/Today/TodayEditionAssigner.swift`
C16	Kairos / novelty	`Sources/Intelligence/Editorial/EditorialNovelty.swift`
C17	Athena / ranking	`Sources/Intelligence/Editorial/EditorialRanking.swift`
C18	Experiment ledger	`docs/TOOLING.md`

The pinned source URLs use the GitHub blob/COMMIT/PATH form. Paths are included as text rather than relying on QR codes or link annotations alone.

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APPENDIX B / SOURCE REGISTER

Code ownership references: C19–C40

Additional sources establish the shared-core, editorial, geopolitical and multi-owner corrections adopted in this edition. C37–C40 were added in edition 15 and C41–C45 in edition 16.

REF	ROLE	PRINTED PATH
C19	Briefing and board	`Sources/News.docc/SituationBoardAndBriefing.md`
C20	Eris / methodology	`docs/methodology/world-tension-v3.md`
C21	Eris / assessment	`Sources/Intelligence/Tension/WorldTension.swift`
C22	Aletheia / quality	`Sources/Intelligence/Overview/OverviewQualityAuditor.swift`
C23	Metis / enrichment	`Sources/Intelligence/Extraction/EnrichmentQueue.swift`
C24	Metis / entity index	`Sources/Intelligence/Editorial/EntityIndexer.swift`
C25	Mnemosyne / entity graph	`Sources/Intelligence/Editorial/EntityGraph.swift`
C26	Clio / reader alignment	`Sources/Intelligence/Editorial/DualReaderAlignmentService.swift`
C27	Clio / perspective reports	`Sources/Intelligence/Editorial/PerspectiveReports.swift`
C28	Iris / wire identity	`Sources/Intelligence/Editorial/EditorialWireIdentity.swift`
C29	Aletheia / prompt defence	`Sources/Intelligence/PromptDefense.swift`
C30	Angelia / model availability	`Sources/Intelligence/AppleIntelligenceAvailability.swift`
C31	Angelia / energy policy	`Sources/Intelligence/IntelligencePolicy.swift`
C32	Athena / Briefing sitrep	`Sources/Intelligence/Briefing/BriefingSitrepComposer.swift`
C33	Mnemosyne / event store	`Sources/Storage/DatabaseEngine+Events.swift`
C34	Clio / generation lifecycle	`Sources/Intelligence/Overview/OverviewGenerationCoordinator.swift`
C35	Architecture boundary	`docs/ARCHITECTURE.md`
C36	Athena / importance	`Sources/Intelligence/Events/StoryImportance.swift`
C37	Iris / feed fetch	`Sources/Services/FeedFetcher.swift`
C38	Metis / act classes	`Sources/Intelligence/Events/ActClass.swift`
C39	Eris / Near You	`Sources/Intelligence/Tension/WorldTensionNearYou.swift`
C40	Today design status	`docs/design/today.md`
C41	Name register	`docs/ANGELIA.md`
C42	Owner-comment check	`script/format.sh`
C43	No single owner / coordinator	`Sources/Coordinators/FeedManager.swift`
C44	Athena / importance audit input	`Sources/Intelligence/Events/EventJudge.swift`
C45	Today page (view, outside Angelia)	`Sources/Views/Today/TodayEdition.swift`

All code references pinned to eb9f73f9205b4c9a32554567eeb1bd96349ea848. These are the implementation owners and paths recorded for the ANGELIA architecture at the pinned commit.

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APPENDIX C / EVIDENCE REGISTER

Measurements, decisions and mythology

Audit files give historical measurements; issue #737 records current research rulings. Classical references explain naming, not software behavior.

ID	EVIDENCE	PRINTED SOURCE PATH
E01	Issue #737: current gate / same-act recall	`issues/737`
E02	10 Nov independent label review	`docs/audits/2026-11-10-label-review.md`
E03	9 Nov evidence-head / upper bound	`docs/audits/2026-11-09-evidence-head.md`
E04	10 Nov veto share out of sample	`docs/audits/2026-11-10-veto-share-out-of-sample.md`
E05	Topic classifier held-out sample	`docs/audits/2026-10-24-topic-classifier-sample.md`
E06	26 Oct perspectives / verifier audit	`docs/audits/2026-10-26-perspectives-and-verifier-controls.md`
E07	6 Nov candidate-language speed	`docs/audits/2026-11-06-candidate-language.md`
E08	6 Nov wire identity	`docs/audits/2026-11-06-editorial-wire-identity.md`
E09	6 Nov novelty evaluation	`docs/audits/2026-11-06-editorial-novelty.md`
E10	6 Nov MMR ranking	`docs/audits/2026-11-06-editorial-ranking.md`
E11	10 Nov Today shadow evaluation	`docs/audits/2026-11-10-today-z2-shadow-evaluation.md`
E12	Synthetic editorial matrix	`docs/audits/2026-11-05-editorial-evaluation.md`
E13	27 Oct Neural Engine placement	`docs/audits/2026-10-27-neural-engine-placement.md`
E14	6 Nov veto + two-member test	`docs/audits/2026-11-06-recall-semantic-veto.md`
E15	6 Nov reclustering stages	`docs/audits/2026-11-06-recluster-stages.md`
E16	27 Oct 3-way pair-model attempt	`docs/audits/2026-10-27-three-way-pair-model.md`
E17	11 Nov Today re-run and comparisons	`docs/audits/2026-11-11-today-z2-comparisons.md`
E18	11 Nov Today page performance	`docs/audits/2026-11-11-today-edition-performance.md`
E19	11 Nov Today first pass in a window	`docs/audits/2026-11-11-today-edition-hardware-pass.md`

Name provenance and literary reading. Angelia: Ἀγγελία, “ang-geh-LEE-ah” (hard g), the personification of messages, listed among Hermes’ children in Pindar. Eris: strife; Clio: muse of history; Mnemosyne: memory and mother of the Muses; Metis: wisdom and counsel, and Athena’s mother in one tradition. Others are job-based analogies, not a shared bloodline.

Myth sources: theoi.com/Daimon/Angelia.html · theoi.com/Olympios/HermesFamily.html · theoi.com/Titan/TitanisMnemosyne.html · theoi.com/Titan/TitanisMetis.html.

Research background: Reimers & Gurevych (2019), aclanthology.org/D19-1410; Reciprocal Rank Fusion (SIGIR 2009), doi:10.1145/1571941.1572114; MMR (SIGIR 1998), doi:10.1145/290941.291025.

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